

Assignment 2: Dual Problem, Gomory Cuts and Column Generation

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1 Problem 1

1.1 Assumptions

Network Flow is as follows:

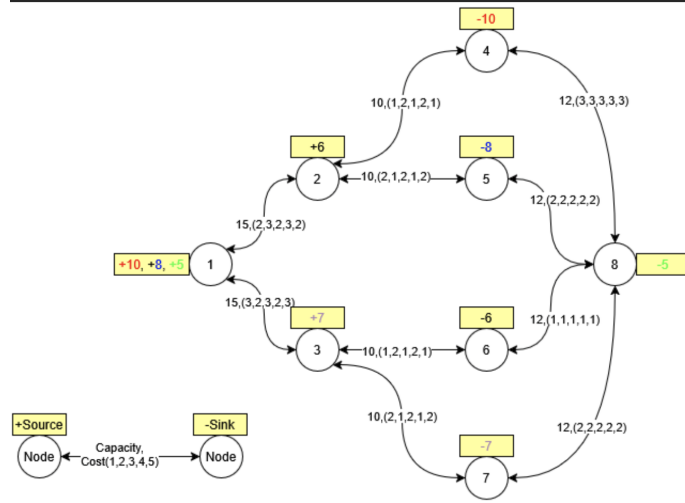


Figure 1: Model of Network Flow

1.2 Given Data

Commodity Flows:

Commodity k	Source $s(k)$	Sink $t(k)$	Demand d_k
1	1	4	10
2	1	5	8
3	2	6	6
4	3	7	7
5	1	8	5

Arc Capacities:

$$u_{1,2} = 15, \quad u_{1,3} = 15$$

$$u_{2,4} = 10, \quad u_{2,5} = 10$$

$$u_{3,6} = 10, \quad u_{3,7} = 10$$

$$u_{4,8} = 12, \quad u_{5,8} = 12$$

$$u_{6,8} = 12, \quad u_{7,8} = 12$$

1.3 Solutions

1.3.1 (a): Multi-Commodity Network Flow Problem (MCNFP)

Sets:

- N : set of nodes (indexed by i or j).
- A : set of (directed) arcs (indexed by (i, j)).
- K : set of commodities (indexed by k).

Parameters:

- c_{ij}^k : cost per unit of commodity k shipped along arc (i, j) .
- u_{ij} : capacity of arc (i, j) .
- d_k : demand (total quantity) of commodity k that must be sent from its source to its destination.
- $s(k)$: source (origin) node of commodity k .
- $t(k)$: sink (destination) node of commodity k .

Decision Variables:

- x_{ij}^k : flow of commodity k on arc (i, j) .

Objective Function:

$$\min \sum_{k \in K} \sum_{(i,j) \in A} c_{ij}^k x_{ij}^k$$

Constraints:

$$\sum_{j:(i,j) \in A} x_{ij}^k - \sum_{j:(j,i) \in A} x_{ji}^k = \begin{cases} d_k, & \text{if } i = s(k), \\ -d_k, & \text{if } i = t(k), \\ 0, & \text{otherwise,} \end{cases} \quad \forall k \in K, \forall i \in N,$$

$$\sum_{k \in K} x_{ij}^k \leq u_{ij}, \quad \forall (i, j) \in A,$$

$$x_{ij}^k \geq 0, \quad \forall k \in K, \forall (i, j) \in A.$$

1.3.2 (b): Associated Dual

Sets:

- N : set of nodes.
- A : set of directed arcs.

- K : set of commodities.

Parameters:

- c_{ij}^k : cost per unit of commodity k on arc (i, j) .
- u_{ij} : capacity of arc (i, j) .
- d_k : demand of commodity k .
- $s(k)$: source nodes for commodity k .
- $t(k)$: sink nodes for commodity k .

Dual Decision Variables:

- π_i^k : dual variable associated with the flow-conservation constraint of commodity k at node i .
- μ_{ij} : dual variable associated with the capacity constraint on arc (i, j) .

Dual Objective Function:

$$\max \sum_{k \in K} \sum_{i \in A} b_i^k \pi_i^k + \sum_{(i,j) \in A} u_{ij} \mu_{ij}.$$

Dual Constraints:

$$\pi_i^k - \pi_j^k + \mu_{ij} \leq c_{ij}^k, \quad \forall k \in K, \forall (i, j) \in A, \quad (1)$$

$$\mu_{ij} \geq 0, \quad \forall (i, j) \in A, \quad (2)$$

$$\pi_i^k \in \mathbb{R}, \quad \forall k \in K, \forall i \in N. \quad (3)$$

1.3.3 (c): Dual Variable / Primal Reduced Cost Relationship

Weak Duality.

- For a maximization problem, any *feasible* primal solution is less than or equal to any *feasible* dual solution.
- For a minimization problem, any *feasible* primal solution is greater than or equal to any *feasible* dual solution.

Unboundedness.

- If the primal is unbounded (for a maximization problem), then the dual is infeasible.
- Conversely, if the dual is unbounded (for a minimization problem), then the primal is infeasible.

Strong Duality.

- If the primal has a finite optimal solution, then the dual also has a finite optimal solution.
- The optimal objective values of the primal and dual coincide at optimality.

Reduced Cost.

- The reduced cost of a variable in the primal indicates how much the objective function would need to improve for that variable to become positive (or “enter the basis” in a simplex context).
- Equivalently, it measures the dual–primal “gap” at the margin of feasibility.

1.3.4 (d): Duality Proof

Let w^* and z^* be the optimal objective values of the (primal) variable set $\mathbf{x}$ and the (dual) variable set $\mathbf{y}$, respectively. Suppose

$$A^T \mathbf{y} \geq \mathbf{c},$$

hence

$$\mathbf{y}^T A \geq \mathbf{c}^T.$$

Because the reduced cost satisfies

$$\mathbf{c}_B^T B^{-1} A - \mathbf{c}^T \geq 0,$$

and $B^{-1} \mathbf{b} = \mathbf{x}_B$ is a feasible solution, we obtain

$$\mathbf{b}^T B^T B^{-1} A^T - \mathbf{b}^T \leq 0.$$

Let

$$w = \mathbf{b}^T \mathbf{y} \quad \text{and} \quad \mathbf{y} = \mathbf{c}_B^T B^{-1}.$$

Then

$$w = \mathbf{b}^T (\mathbf{c}_B^T B^{-1})^T = \mathbf{b}^T (B^{-1})^T \mathbf{c}_B.$$

Noting that $(B^{-1})^T \mathbf{c}_B = B^{-1} \mathbf{c}_B$ when vectors and matrices have the appropriate dimensions, we get

$$\mathbf{b}^T (B^{-1})^T \mathbf{c}_B = \mathbf{c}_B^T (B^{-1} \mathbf{b}) = \mathbf{c}_B^T \mathbf{x}_B = z.$$

Thus,

$$w = z,$$

and so at optimality, the primal and dual objective values coincide.

1.3.5 (e): Complementary Slackness

Complementary slackness expresses the relationship between optimal primal and dual solutions. It requires that for each primal constraint, either the corresponding dual variable is zero or the primal constraint is active, and vice versa. These conditions help verify the optimal solutions in both the primal and dual formulations.

2 Problem 2

2.1 Solutions

2.1.1 (a): MILP Formulation

Sets:

- S : set of stations (indexed by i or j).
- $K = \{(i, j) \mid i, j \in S, i \neq j\}$: set of OD pairs.
- R : set of routes (indexed by r).

Parameters:

- D_{ij} : demand from station i to station j .
- $t_{(i,j),r}$: travel time for OD pair (i, j) on route r .
- c_r : cost of route r .
- C : capacity of a bus.
- B : maximum number of buses per route.
- T : total travel-time budget.

Decision Variables:

- $x_{(i,j),r} \geq 0$: flow of OD pair (i, j) on route r .
- $y_r \in \{0, 1\}$: binary indicator for using route r .
- $b_r \in \{0, 1, \dots, B\}$: number of buses on route r .

Objective Function:

$$\min \sum_{r \in R} c_r y_r$$

Constraints:

$$\sum_{r \in R} x_{(i,j),r} = D_{ij} \quad \forall (i, j) \in K,$$

$$\sum_{(i,j) \in K} x_{(i,j),r} \leq C b_r \quad \forall r \in R,$$

$$b_r \leq B y_r \quad \forall r \in R,$$

$$\sum_{(i,j) \in K} \sum_{r \in R} t_{(i,j),r} x_{(i,j),r} \leq T,$$

$$x_{(i,j),r} = 0 \quad \text{if} \quad t_{(i,j),r} = 0.$$

2.1.2 (b): Gurobi Solution

Objective (min cost) = 21600.0

Number of routes used = 16

Total passenger travel time = 333080.0 (limit = 391890)

Route Index	Stations	y	Buses	Route Cost
18	(1, 2, 3, 5)	1.0	5.0	1400
65	(1, 3, 4)	1.0	5.0	1300
90	(1, 3, 6, 7)	1.0	5.0	1400
94	(1, 3, 7, 8)	1.0	5.0	1400
172	(2, 4, 6, 8)	1.0	5.0	1400
173	(2, 4, 7)	1.0	5.0	1300
233	(4, 5, 6)	1.0	5.0	1300
267	(5, 7, 8)	1.0	5.0	1300
273	(6, 3, 1)	1.0	5.0	1300
292	(6, 5, 4)	1.0	5.0	1300
312	(7, 4, 2)	1.0	5.0	1300
324	(7, 5, 3, 2)	1.0	5.0	1400
369	(8, 2, 1)	1.0	5.0	1300
379	(8, 4, 3, 1)	1.0	5.0	1400
447	(8, 7, 5, 1)	1.0	5.0	1400
464	(8, 7, 6, 2)	1.0	5.0	1400

Table 1: Summary of selected routes and their usage.

The optimization program in Gurobi was terminated at 600 seconds. This was later repeated at 3600 seconds, with the same reported results.

3 Problem 3

3.1 Assumptions

As a reasonable assumption to the problem, considering integer levels of production, values for maximum demand and minimum production are preprocessed to eliminate trivial solutions. Floor values are used in the case of maximum demand. Ceiling values are used in the case of minimum production. This eliminates fractional values between the given constraints and the feasible integer space. These may be considered Heuristic cuts.

3.2 Solutions

3.2.1 (a): LP Relaxation

Sets:

- P : set of products.

Parameters:

- c_p : profit per unit of product $p \in P$.
- t_p : time required per unit of product $p \in P$.
- d_p : maximum demand of product $p \in P$.
- b_p : minimum production required of product $p \in P$.
- L : limit on total production time.

Decision Variables:

- x_p : units produced of product $p \in P$

Objective Function:

$$\max \sum_{p \in P} c_p x_p$$

Constraints:

$$x_p \leq d_p, \quad \forall p \in P,$$

$$x_p \geq b_p, \quad \forall p \in P,$$

$$\sum_{p \in P} t_p x_p \leq L,$$

$$x_p \geq 0, \quad \forall p \in P.$$

3.2.2 (b): Branch and Bound

Initial feasible solution from the relaxed LP atop the branch and bound tree.

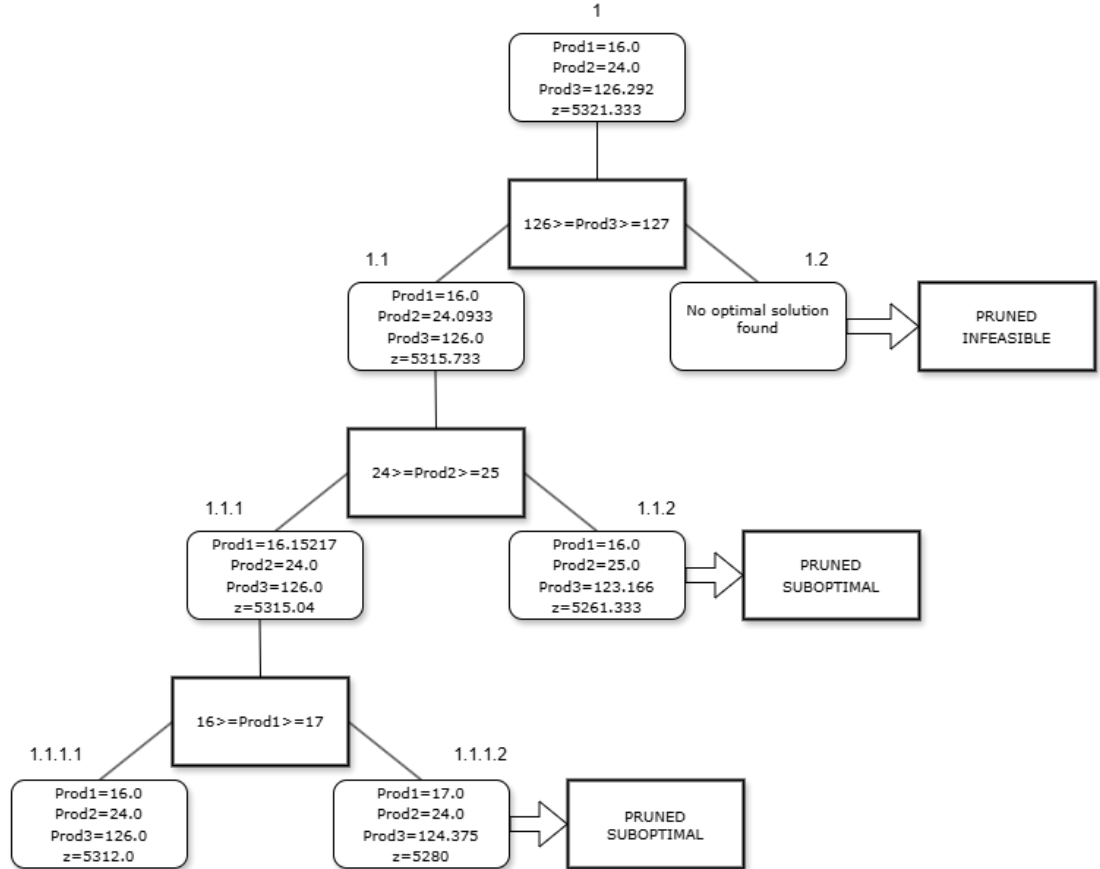


Figure 2: Branch and Bound Tree with Actions Descriptions

With INTEGER Solution found where Prod1= 16.0, Prod2= 24.0, Prod3= 126.0

The value of the objective function is 5312.0 at the indicated production level.

3.2.3 (c): Search Differences

Depth-First Search.

- This strategy first explores one branch until a complete feasible integer solution is found, then backtracks to identify if any other branches produce more optimal solutions.

Best-Node-First Search.

- Best node first selects the node with the highest LP relaxation bound to explore next and iterates with this same preference until optimality is obtained.

Least-Discrepancy Search.

- Follows a heuristic ordering that is suggested by the LP solution, first exploring nodes that deviate the least from the LP solution and continuing there-on.

3.2.4 (d): Branch and Cut Explanation

Cutting planes are inequalities added to the LP relaxation that remove fractional solutions without excluding the feasible integer solutions. Cutting planes tighten the relaxed LP, which improves the bound and can reduce the number of nodes in the branch and bound tree.

3.2.5 (e): Branch and Cut Implementation